Escape from Typicality: Why Evolution Operates Near One Mutation per Genome per Generation

Ishanu Chattopadhyay[a]

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Across organisms and viruses, the per-site mutation rate scales inversely with genome length, yielding an approximately constant number of mutations per genome per generation (Drake's rule). Existing explanations emphasize biochemical fidelity constraints or population-genetic error-threshold arguments that limit mutation from above. Here we provide a complementary, mechanism-independent account based on variation supply. Treating mutation as a blind local perturbation process, we ask which mutation rates maximize the expected rate of producing variants that are atypical relative to the mutation kernel itself, and thus supplied at elevated rate relative to the mutational background, and therefore available for selective amplification. Using counting bounds for the fraction of such mutation-atypical outcomes within Hamming neighborhoods and combining them with a standard local mutation model (e.g., independent per-site substitutions), we obtain an explicit discovery-rate expression that is maximized when the per-genome mutation intensity $T = n\mu$ is $O(1)$, with a peak near $T \approx 1$. Consequently, the optimal per-site rate scales as $\mu^\star = (1 + o(1))/n$. These results derive Drake's rule from a generic supply–destruction tradeoff under local stochastic perturbations, complementing fidelity- and stability-based perspectives.

biological evolution | mutation rate | algorithmic complexity | Drake's rule

Mutation injects randomness; selection preserves and amplifies structure. Evolution therefore operates under a fundamental tension between entropy production and entropy suppression. A striking empirical regularity across diverse taxa is that the per-site mutation rate μ scales approximately as $1/n$, where n is genome length, so the per-genome mutation rate $U = n\mu$ is roughly constant. This phenomenon, observed from RNA viruses to eukaryotes, is known as Drake's rule (1–4).

Standard explanations of Drake's rule fall into two broad categories. One emphasizes biochemical constraints and tradeoffs in replication fidelity, proofreading, and energetic cost (2). The other emphasizes the population-genetic consequences of excessive mutation, most prominently Eigen's error-threshold framework (5), in which high mutation loads destabilize inherited information and limit the maintenance of adapted structure. These approaches provide important insight and useful upper bounds, but they primarily explain why mutation rates cannot be too large. They do not, by themselves, explain why evolution repeatedly operates near a particular scaling regime with $U = O(1)$.

Here we pursue a complementary route that is upstream of selection and does not invoke biochemical detail, explicit fitness landscapes, or equilibrium population genetics. We ask: given a blind local mutation mechanism, at what mutation load is the *supply rate* of statistically exceptional variants maximized? The point is not that evolution seeks exceptionality per se, but that selection can only act on what mutation supplies, namely for a parent genome x:

$$x \xrightarrow{\text{mutation}} \text{Large set of random mutants} \xrightarrow{\text{selection}} \{y_1, \ldots, y_k\} \qquad [1]$$

If mutation predominantly produces outcomes that are typical under its own kernel, then repeated mutation pushes lineages toward the kernel's typical set, and accumulated organization is continually eroded into mutational background; we descend into noise (See Fig.1a). Any sustained adaptive process therefore requires a nonzero supply of outcomes that are atypical relative to mutation alone, since these are the only candidates that can be preferentially retained rather than washed out. Thus, we are not proposing any explicit evolutionary objective function; we are analyzing structural constraints on variation supply induced by local blind search. This perspective also connects naturally to evolvability, the capacity of populations to generate heritable variation that can support adaptive change (6, 7). Mutation rates are themselves evolving traits shaped by indirect selection through their

Author affiliations: [a]University of Kentucky

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[2]To whom correspondence should be addressed. E-mail: ishanu_ch@uky.edu

effects on future variation (8, 9). Since mutation supplies the heritable novelty on which selection can act, rate-tuning can arise naturally under slection pressure to converge over time to offer maximal discovery advantage.

Our starting observation is that mutation induces a natural statistical model for an offspring relative to its parent. For a parent genome $x \in [q]^n$, an m-mutation restricts the offspring y to the Hamming sphere $H_m(x)$, the set of sequences at Hamming distance m from x. Under the mutation-induced model that is uniform over $H_m(x)$, a typical draw requires $\log |H_m(x)|$ bits to specify once the radius is fixed, and $|H_m(x)| = \binom{n}{m}(q-1)^m$ grows rapidly with m. This immediately implies a notion of typicality: at a given radius, most outcomes resemble typical draws from the mutation kernel, and only a small fraction are exceptional relative to that kernel. In particular, the larger the neighborhood, the thinner the tail of mutation-atypical outcomes within it.

We formalize this atypicality or exceptionality relative to the mutation-induced model on $H_m(x)$ using randomness deficiency (Defn. S1), which measures how much rarer an outcome is than a typical draw from the same model. Rarity here is not a proxy for fitness; rather, being rare relative to the mutation kernel is what makes an outcome statistically distinguishable from mutational background and therefore *available*, in principle, for selective amplification. A mutant is called a *net structure discovery* if its deficiency exceeds a margin Δ (Def. S1). A two-part coding/counting argument (Thm. S1, Lem. S1) implies that, for typical parents, the fraction of Δ-exceptional outcomes at radius m scales as $2^{-\Delta}/|H_m(x)|$ up to polylogarithmic factors; this "typical" qualifier is in a counting/combinatorial sense (formalized in the SI Appendix) and is not an empirical claim that biological genomes are algorithmically random. (However, real genomes often lie close to the maximal entropy rate for a four-letter alphabet, consistent with the combinatorial typicality regime, see Fig.1b.) We then combine this typical-case rarity with an independent per-site mutation model $M \sim \text{Binomial}(n, \mu)$ to define an expected discovery rate $\Phi(\mu)$ that averages the Δ-exceptional probability over mutation radii (Eq. (S14)). This choice reflects independent per-site mutation as a null local-perturbation model; the scaling result however depends only on thin-tailed locality, not on strict independence. Optimizing $\Phi(\mu)$ yields an interior optimum at per-genome mutation intensity $T = n\mu = O(1)$ (Theorem S2), and in particular,

$$\mu^\star = \frac{1 + o(1)}{n}.$$

Thus Drake's rule emerges as a mechanism-independent supply optimum under blind local perturbation, complementing fidelity- and stability-based perspectives such as the error-threshold viewpoint.

Discussion

This work provides a mechanism-independent route to Drake's rule from the standpoint of variation supply. Under a minimal model in which mutation acts as a local stochastic perturbation on a discrete sequence space, we derived a typical-case bound on the probability that an m-mutant is statistically exceptional relative to its mutation neighborhood. Combining this rarity scaling with an independent per-site mutation model yields an explicit expression for the expected

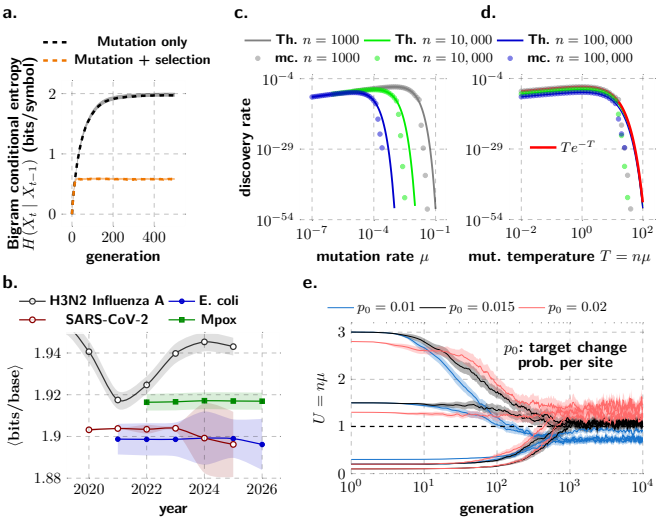

Fig. 1. Scaling and collapse of the discovery rate under independent per-site mutation. a. Toy evolution on $[4]^n$ under mutation alone versus mutation with a viability filter that accepts offspring only within a fixed Hamming distance of a structured template. The plotted statistic is the empirical bigram conditional entropy $H(X_t \mid X_{t-1})$ (bits per symbol) along replicate lineages; mutation alone drives trajectories toward the maximal entropy regime for a four-letter alphabet, while the viability filter maintains bounded entropy. **b.** Genome compression values over time (viral genomes from GISAID; *E. coli* from NCBI; see SI) illustrating empirical boundedness of sequence complexity over evolution and the near-incompressibility of genomes under a reference-free compression proxy. The Shannon ceiling for a four-letter alphabet is 2 bits/base, achieved by perfectly incompressible sequences. Shaded bands indicate 95% confidence intervals across genomes within each year. **c.** The expected discovery rate $\Phi(\mu)$ as a function of the per-site mutation rate μ for increasing genome lengths n (Th. = theory; mc. = Monte Carlo estimates). The maximizer shifts to smaller μ as n increases, consistent with the inverse-length scaling $\mu^\star \propto 1/n$. **d.** The same discovery rate plotted against the per-genome mutation intensity $T = n\mu$. Curves for different n collapse onto a common profile with a maximum near $T \approx 1$; we illustrate the predicted Te^{-T} dependence up to a multiplicative constant. The collapse demonstrates that the optimal regime corresponds to an $O(1)$ number of mutations per genome per generation. **e.** Simplified binary genome simulation with heritable mutation-rate modifiers under a moving target. Across diverse initial conditions populations evolve toward an intermediate mutation intensity, consistent with indirect selection on mutation-rate modifiers under sustained demand for adaptive tracking.

rate at which mutation supplies such exceptional variants, and shows that this rate is maximized when the expected number of mutations per genome per generation is $O(1)$. In the usual per-site parametrization, this corresponds to the inverse-length scaling $\mu^\star = \Theta(1/n)$.

A useful way to express the optimum is through the per-genome mutation intensity

$$T := n\mu = \mathbb{E}[M], \qquad M \sim \text{Binomial}(n, \mu),$$

which acts as an intensive "mutation temperature" controlling the typical perturbation size. Figure 1 illustrates the scaling directly. In Fig. 1c, $\Phi(\mu)$ plotted against the per-site rate μ exhibits a single interior maximum whose location shifts left as n increases, consistent with $\mu^\star \propto 1/n$. In Fig. 1d, plotting the same discovery rate against $T = n\mu$ collapses curves for different n onto a common profile with a maximum near $T \approx 1$, matching the predicted Te^{-T} dependence up to a multiplicative constant. The collapse emphasizes that the scaling law is most naturally stated as a constant optimal per-genome perturbation size. More precisely, the exact finite-n expression involves the factor $(1-\mu)^n$, so that the leading behavior is proportional to $T(1-\mu)^n$; the Te^{-T}

profile is its small-μ asymptotic form. The existence of an interior optimum at $T = O(1)$ does not depend on the Poisson approximation itself but on the independence and thin-tailed nature of the mutation count distribution.

Our notion of discovery is defined relative to the distribution induced by the mutation mechanism itself. This reference measure is not an external modeling choice but is canonically determined by the perturbation process that supplies variation. Selection can only act on outcomes drawn from this kernel; thus atypicality must be measured relative to it. We do not claim that the Hamming sphere is the globally optimal statistical model for an offspring, only that it is the model canonically induced by a local mutation mechanism on sequence space. The substantive claim is upstream: since mutation is the sole source of variation, selection can only act on what mutation supplies. If offspring were always typical draws from the mutation kernel, repeated perturbation would concentrate lineages toward the kernel's typical set, eroding accumulated organization into mutational background. Positive randomness deficiency provides a quantitative, mechanism-independent way to express the minimal sense in which a variant is not merely background mutational noise.

The inverse-length scaling is robust to the novelty margin Δ, provided Δ does not grow with n fast enough to dominate the one-step indexing cost $c(1) = \Theta(\log n)$. In particular, any fixed Δ or $\Delta = o(\log n)$ yields the same $\mu^\star = \Theta(1/n)$ scaling, with Δ affecting only the overall rate via the prefactor $2^{-\Delta}$.

To connect this perspective to evolvability, we carry out a simple simulation experiment in Fig. 1e to illustrate how an order-one mutation regime can become a target of indirect selection rather than a built-in objective. Each individual genome (binary sequence) carries its own mutation rate drawn from a distribution, and offspring viability is defined by Hamming distance to a moving target sequence whose sites change independently with probability p_0. Across a wide range of p_0 values and initial mutation rates, populations evolve toward an intermediate genomic mutation intensity in the order-one range. Low mutation rates are disfavored because they under-supply adaptive variants under environmental change, whereas high mutation rates generate excess low-fitness offspring and erode accumulated structure. Despite being highly stylized, this experiment shows that selection on evolving mutation rates can favor regimes that sustain a reliable supply of adaptive variation (8, 10). Population-genetic accounts such as drift-barrier arguments (9) remain complementary: they constrain how far fidelity can improve in finite populations, whereas the framework presented here identifies which mutation regime most effectively supplies mutation-atypical variants once mutation is treated as a blind local generative process.

Why focus on discovery probability?. Why should evolution optimize the probability of structure discovery rather than a more direct quantity such as expected fitness gain (2–4)? Mutation has no access to fitness: it is a blind local perturbation mechanism that generates variants without foresight, and selection acts only after these variants have been produced. Any objective defined in terms of fitness therefore lives downstream of mutation and cannot be optimized at the level of the generative process itself. The discovery probability instead isolates an upstream constraint imposed by mutation on the supply of selectable variants.

It is also useful to situate this result relative to classical "error threshold" arguments. Eigen-type frameworks constrain mutation from above by requiring that inherited information remain stable under copying errors (5). Our result is different in kind: it identifies an interior optimum for the upstream *supply* of mutation-atypical variants under blind local perturbations, without invoking a fitness landscape or equilibrium population genetics. The fact that both perspectives emphasize the $U = T = n\mu = O(1)$ regime suggests that Drake's rule may reflect the intersection of two generic pressures: fidelity constraints that limit loss of existing information and supply constraints that limit how often mutation produces variants sufficiently atypical to be preferentially amplified. In this sense, operating near one mutation per genome per generation can be read as a regime in which both retention and exploration remain nontrivial.

If the perturbation kernel becomes strongly non-local (e.g., recombination, large indels) or highly overdispersed in mutation count (*e.g.*, mutator mixtures or burst-like replication errors (3, 11)), the detailed functional form of $\Phi(T)$ can change and the maximizing value may shift. However, for any local, thin-tailed perturbation kernel parameterized by an intensive scale T, the supply/erosion tradeoff generically produces an interior optimum at order-one T.

Beyond biology. Although motivated by biological evolution, the derivation depends only on three ingredients: a discrete configuration space, a local stochastic perturbation mechanism, and the requirement that adaptive progress requires access to outcomes that are atypical relative to the perturbation mechanism itself. Any system that explores a high-dimensional space via local random perturbations (a blind search unaware of downstream fitness functions) faces an analogous tradeoff: perturbations that are too small rarely generate statistically exceptional outcomes, while perturbations that are too large overwhelm any low-complexity deviation by pushing outcomes into exponentially large neighborhoods in which the probability of exceptionality is sharply suppressed. The resulting optimum selects an $O(1)$ intensive perturbation scale, made visible here by the collapse of discovery-rate curves under the parameter T in Fig. 1d.

In summary, the combinatorial growth of mutation neighborhoods suppresses the probability of mutation-atypical outcomes at large radii, and optimizing the resulting discovery rate under independent per-site mutation yields the observed inverse scaling $\mu^\star \sim 1/n$, equivalently an $O(1)$ mutation load per genome.

Materials and Methods

Approach. We model mutation as a blind local perturbation on $[q]^n$: conditional on m mutated sites, offspring are drawn uniformly from the Hamming sphere $H_m(x)$ around a parent x (See Supplementary Methods for formal framing). For any finite model class $S \ni y$, the two-part code bound gives

$$K(y) \;\le\; K(S) + \log|S| + O(1), \qquad [2]$$

so describing y via S incurs an intrinsic indexing cost $\log|S|$. Applying this with $S = H_m(x)$ yields

$$K(y) \;\approx\; K(x) + \log|H_m(x)|, \qquad |H_m(x)| = \binom{n}{m}(q-1)^m, \quad [3]$$

so the combinatorial growth of the mutation neighborhood determines the typical description length of an m-mutant (See Eq. (S2) - Eq. (S3)). We quantify mutation-atypicality using randomness deficiency (Eq. (S5)), which measures how much rarer an outcome is than a typical draw from the mutation-induced model. A mutant is called a *net structure discovery* if its deficiency exceeds a margin Δ (Def. S1). A counting argument (Thm. S1, Lem. S1) shows that only a $2^{-\Delta}/|H_m(x)|$ fraction of outcomes at radius m can be Δ-exceptional, up to polylogarithmic factors.

Using this local rarity within an independent per-site mutation model $M \sim \text{Binomial}(n, \mu)$, and averaging over mutation radii yields an expected discovery rate $\Phi(\mu)$ (Cor. S1). Analyzing the small-μ asymptotics of $\Phi(\mu)$ (Lem. S2) and optimizing with respect to μ (Thm. S2) yields a $n\mu = O(1)$ optimum (Eq. (S20)).

Code availability. Instructions and code to generate Fig. 1 panels are available at https://doi.org/10.5281/zenodo.18676492

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